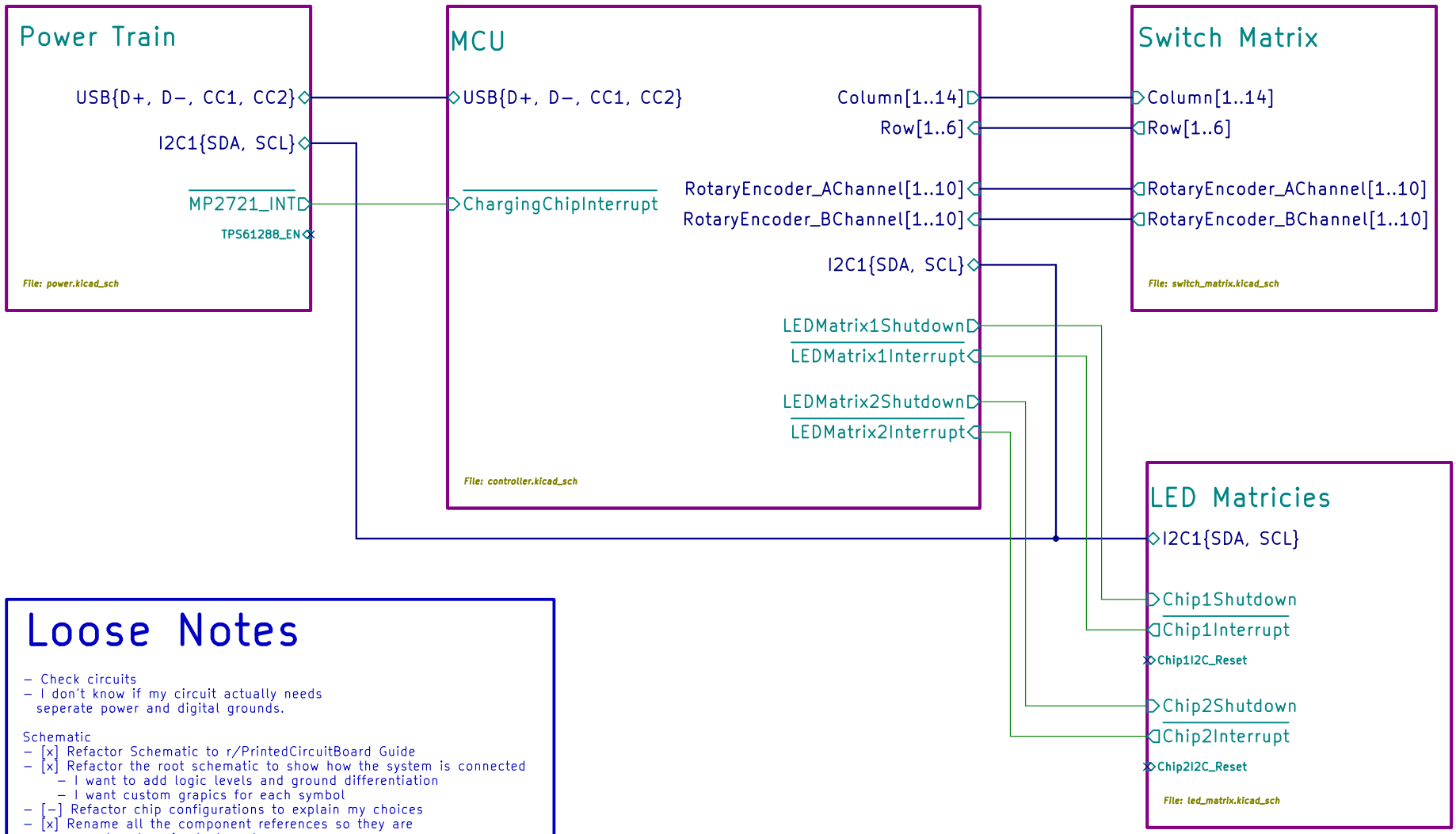


Loose Notes

- Check circuits
- I don't know if my circuit actually needs seperate power and digital grounds.

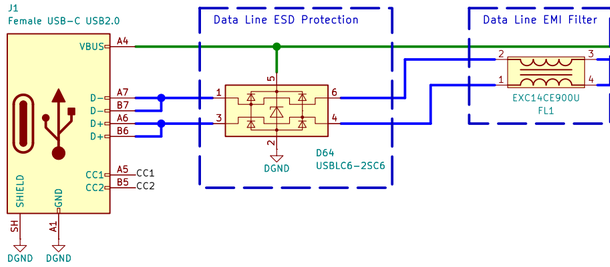
Schematic
- [x] Refactor Schematic to r/PrintedCircuitBoard Guide
- [x] Refactor the root schematic to show how the system is connected
  - I want to add logic levels and ground differentiation
  - I want custom grapics for each symbol
- [-] Refactor chip configurations to explain my choices
- [x] Rename all the component references so they are grouped and easier to layout
- [x] Select actual components, footprints, 3D models for all components
- [x] Pass those design reviews softwares
- [x] take review advice

PCB
- generate and recreate ergonomic montary key layout in PCB
- determine latch layout on PCB

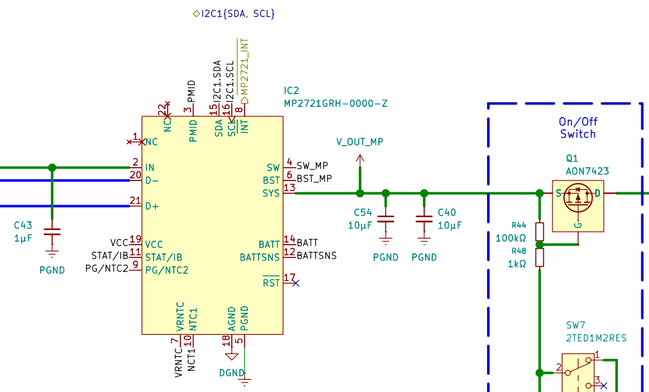
KiCad Feature Requests:
Hierarchical Bus
Hierarchical Sheet Symbols

Sheet: /		
File: slack.kicad_sch		
Title: Slack		
Size: A	Date: 2026-07-25	**Rev: A.0.5**
KiCad E.D.A. 10.0.5		Id: 1/7

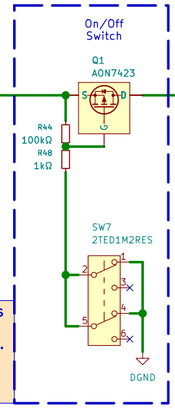
USB-C 2.0



Charger

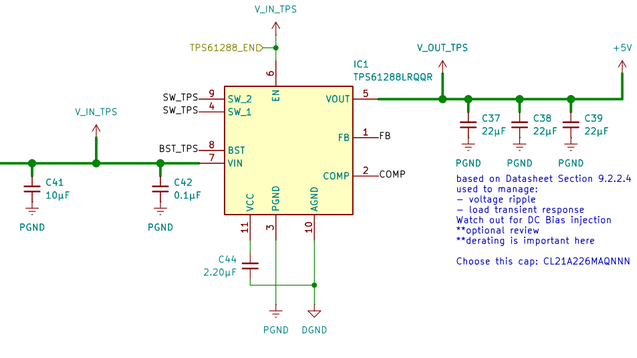


Few desirable switches are rated for max current of this system. So we're using a load switch MOSFET.



Boost Converter

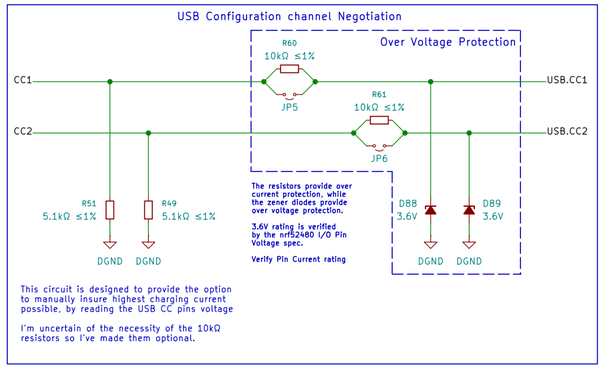
*set to always on. trying to control this would be a power squencing nightmare

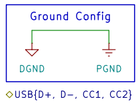


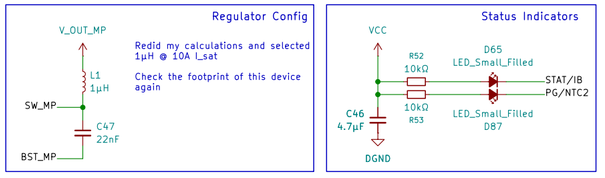
based on Datasheet Section 9.2.2.4 used to manage:
- voltage ripple
- load transient response

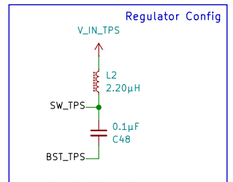
Watch out for DC Bias injection
**optional review
**derating is important here

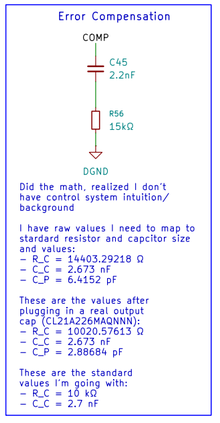
Choose this cap: CL21A226MAQNNN

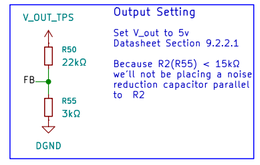


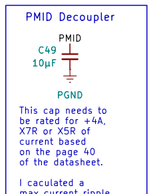


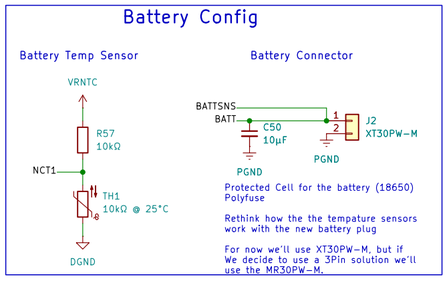












Do I have enough protect on the USB contector?
Do I still need to add resistors or pull up these lines?
Have I properly set up the charger and regulator correctly?
why can charge current be higher than input current in battery charger
Zener Diodes vs Schottky Diode for pin protection

Power Ratings per Critical Line
USB-C plug → Charger IC input:
- max 5V @ 3A
- must be configured

USB-C plug → Charger IC output:
- Worst case output current draw 6A
- Max output current 5A
- Voltage range 3.125V ↔ 3.913V
  default 3.7V

Battery → Charger IC output:
- Worst case discharge current draw 7.5A
- Max discharge current 8A
- Battery Voltage 3V ↔ 4.2V

Voltage Regulator: Rating 5A, 5.5V
- max 5V @ 4A

FIRMWARE NOTE:
Power scenarios must be accounted for in the firware.

Selection of passives:
resistors, capcitors, inductors, diodes, switchs, LEDs, usb port

Consider BQ25640 & BQ25895

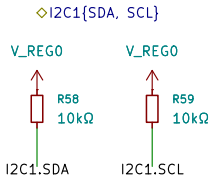
Sheet: /Power Train/
File: power.kicad_sch

Title: Slack

Size: User	Date: 2026-07-25	**Rev: A.0.5**
KiCad E.D.A. 10.0.5		Id: 2/7

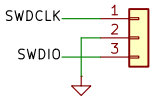
Microcontroller

I2C Bus Config



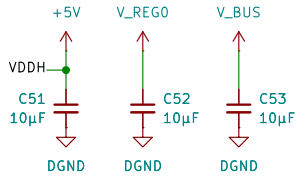
pick theses last based on the capacitance of the line on the PCB

Programmer



Verify that this is enough to program this board

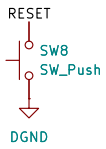
Power Config



Regulator 0 & Regulator 1 are both set to LDO mode

VDD is REG0, it defaults to +1V8 and can be set to V_REG0.

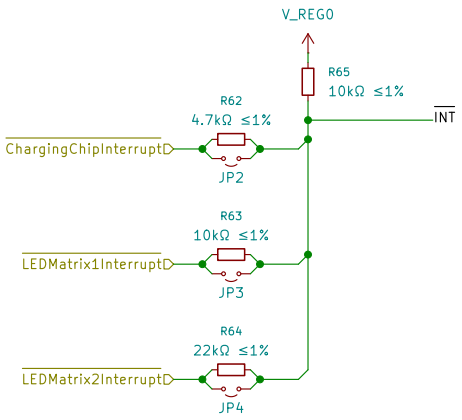
Reset Button



The datasheet in unclear about behavior of this pin

I't is possible we may need to set this manually

Interrupt Multiplexing



This circuit is designed to allow for both a analog and digital method of interrupt detection.

It uses a voltage ladder.
All values varified

Active interrupts	Node voltage
none	3.30 V
driver #2 (22 k)	2.27 V
driver #1 (10 k)	1.65 V
#1 + #2	1.34 V
MP2721 (4.7 k)	1.06 V
MP2721 + #2	0.92 V
MP2721 + #1	0.80 V
all three	0.72 V

Loose Notes

Low Frequency IO Pins:
= P0.02 P0.03 P0.09 P0.10 P0.28 P0.29 P0.30 P0.31
= P1.01 P1.02 P1.03 P1.04 P1.05 P1.06 P1.07 P1.10
= P1.11 P1.12 P1.13 P1.14 P1.15

Standard Frequency IO Pins:
= P0.00 P0.01 P0.04 P0.05 P0.06 P0.07 P0.08 P0.11
= P0.12 P0.13 P0.14 P0.15 P0.16 P0.17 P0.18 P0.19
= P0.20 P0.21 P0.22 P0.23 P0.24 P0.25 P0.26 P0.27
= P1.00 P1.08 P1.09

P0.18 is the nRESET Pin. The datasheet is unclear about how reset works

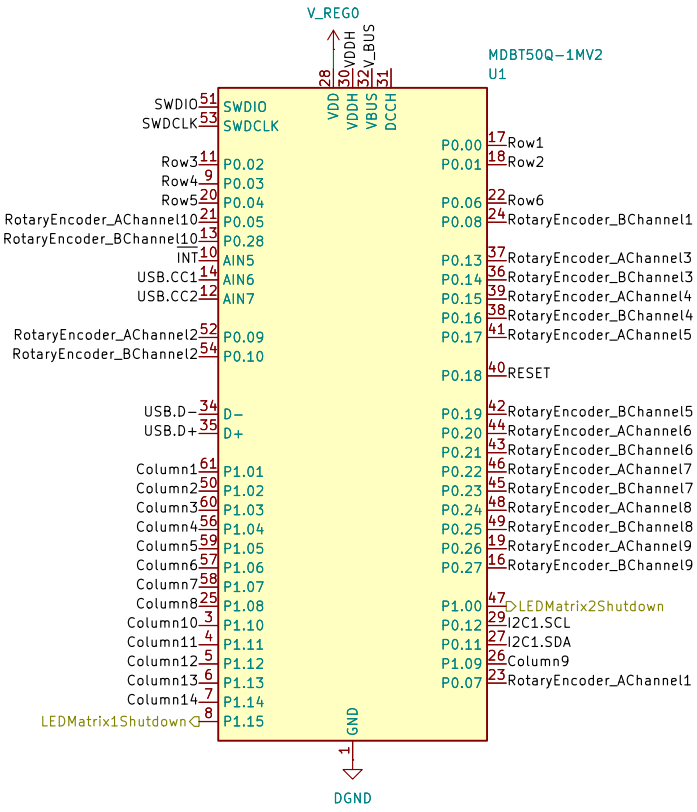
Be careful with the pin assignments as not every pin can handle a high speed signal

Almost any pin can be used for i2c on the nrf52. Sight this source
So I'll use P0.00 and P0.01.
This might be problematic as these pins are used for the secondary clock.

Consider if there enough space for UART debug. No there will not be

Set up resets for the device.
Centralize the decoupling for VDDH

Column[1..14]
Row[1..6]
RotaryEncoder_AChannel[1..10]
RotaryEncoder_BChannel[1..10]
USB{D+, D−, CC1, CC2}



Sheet: /MCU/		
File: controller.kicad_sch		
Title: Slack		
Size: User	Date: 2026-07-25	**Rev: A.0.5**
KiCad E.D.A. 10.0.5		Id: 3/7

Main Matrix

Column[1..14]
Row[1..6]
RotaryEncoder_AChannel[1..10]
RotaryEncoder_BChannel[1..10]

Row1
Row2
Row3
Row4
Row5
Row6

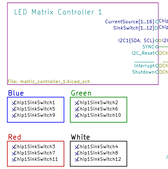
Latch Row

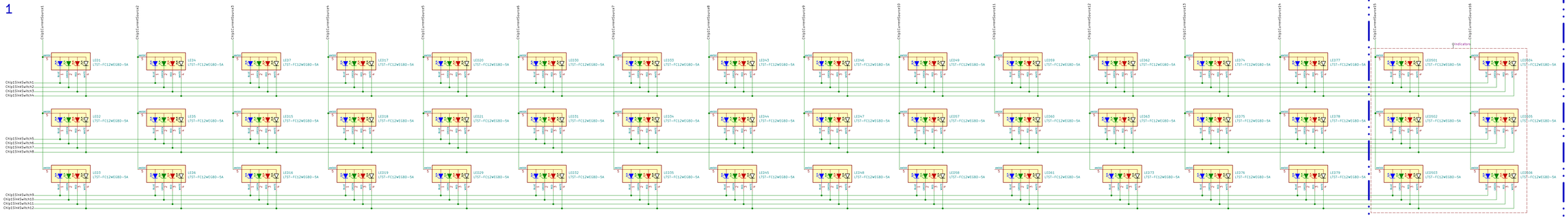
RotaryEncoder_AChannel1
RotaryEncoder_BChannel1
RotaryEncoder_AChannel2
RotaryEncoder_BChannel2
RotaryEncoder_AChannel3
RotaryEncoder_BChannel3
RotaryEncoder_AChannel4
RotaryEncoder_BChannel4
RotaryEncoder_AChannel5
RotaryEncoder_BChannel5
RotaryEncoder_AChannel6
RotaryEncoder_BChannel6
RotaryEncoder_AChannel7
RotaryEncoder_BChannel7
RotaryEncoder_AChannel8
RotaryEncoder_BChannel8
RotaryEncoder_AChannel9
RotaryEncoder_BChannel9
RotaryEncoder_AChannel10
RotaryEncoder_BChannel10

V_REGO
DGND

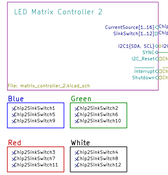
RE1/SW7 RotaryEncoder_Switch
RE2/SW8 RotaryEncoder_Switch
RE3/SW21 RotaryEncoder_Switch
RE4/SW22 RotaryEncoder_Switch
RE5/SW35 RotaryEncoder_Switch
RE6/SW36 RotaryEncoder_Switch
RE7/SW49 RotaryEncoder_Switch
RE8/SW50 RotaryEncoder_Switch
RE9 RotaryEncoder
RE10 RotaryEncoder

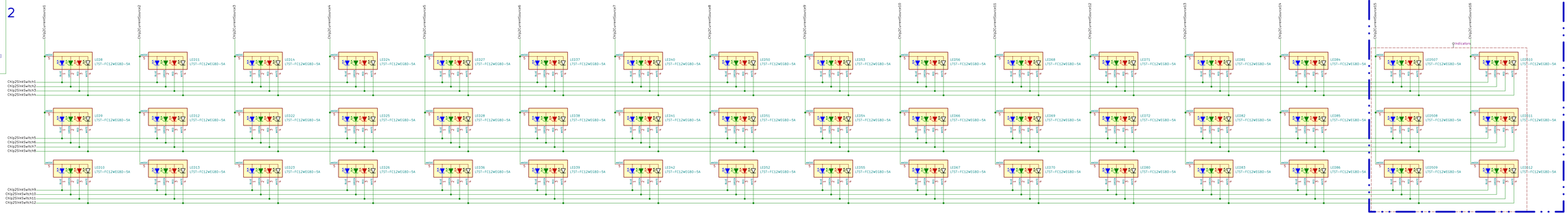
RGBW LED Matrix Halve 1





RGBW LED Matrix Halve 2



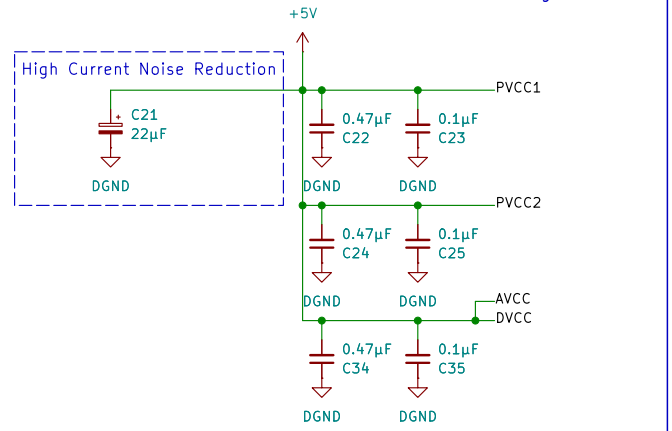


Indicators

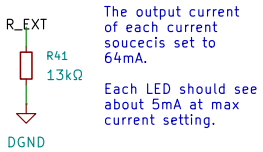
Sheet: /LED Matrices/		
File: led_matrix.kicad_sch		
Title: Slack		
Size: User	Date: 2026-07-25	**Rev: A.0.5**
KiCad E.D.A. 10.0.5		Id: 5/7

LED Matric Controller 1

Power Config



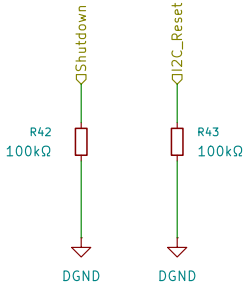
LED max current setting



The output current of each current soucecis set to 64mA.

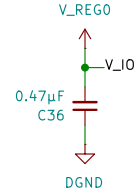
Each LED should see about 5mA at max current setting.

Signal Config

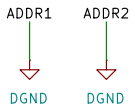


I2C Config

logic reference voltage



I2C address setting



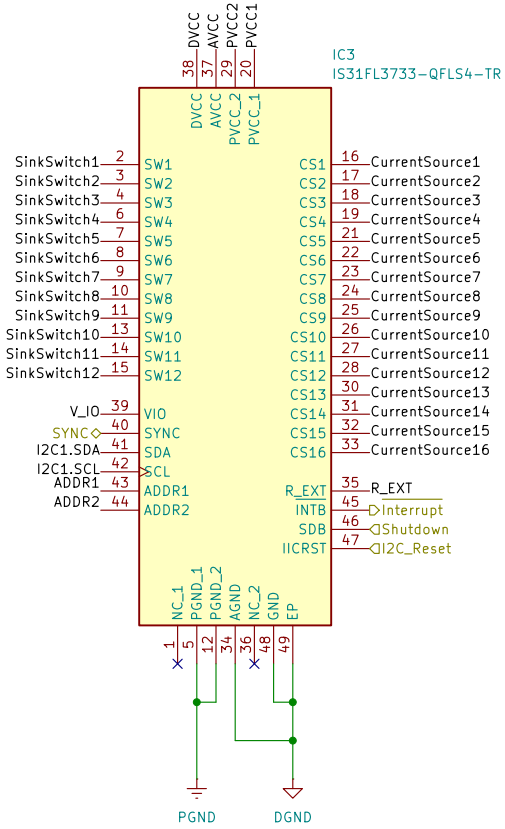
CurrentSource[1..16]
SinkSwitch[1..12]
I2C1{SDA, SCL}
SYNC
I2C_Reset
Shutdown
Interrupt

Blue
- SinkSwitch1
- SinkSwitch5
- SinkSwitch9

Green
- SinkSwitch2
- SinkSwitch6
- SinkSwitch10

Red
- SinkSwitch3
- SinkSwitch7
- SinkSwitch11

White
- SinkSwitch4
- SinkSwitch8
- SinkSwitch12



*is the use of 2 seperate grounds necessary for this?

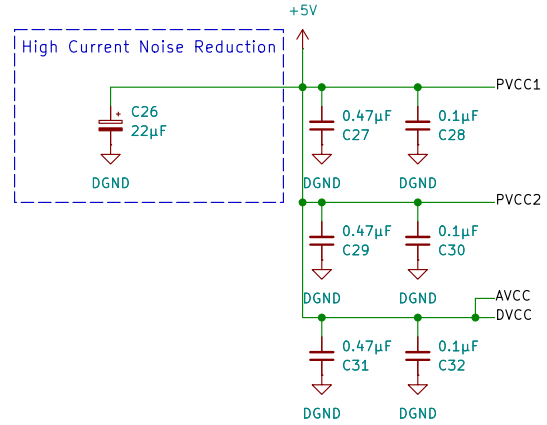
Sheet: /LED Matricies/LED Matrix Controller 1/
File: matric_controller_1.kicad_sch

Title: Slack

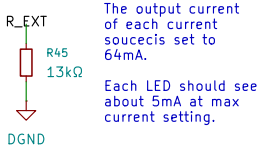
Size: A	Date: 2026-07-25	**Rev: A.0.5**
KiCad E.D.A. 10.0.5		Id: 6/7

LED Matric Controller 1

Power Config



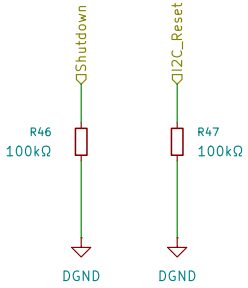
LED max current setting



The output current of each current soucecis set to 64mA.

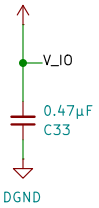
Each LED should see about 5mA at max current setting.

Signal Config



I2C Config

logic reference voltage



I2C address setting



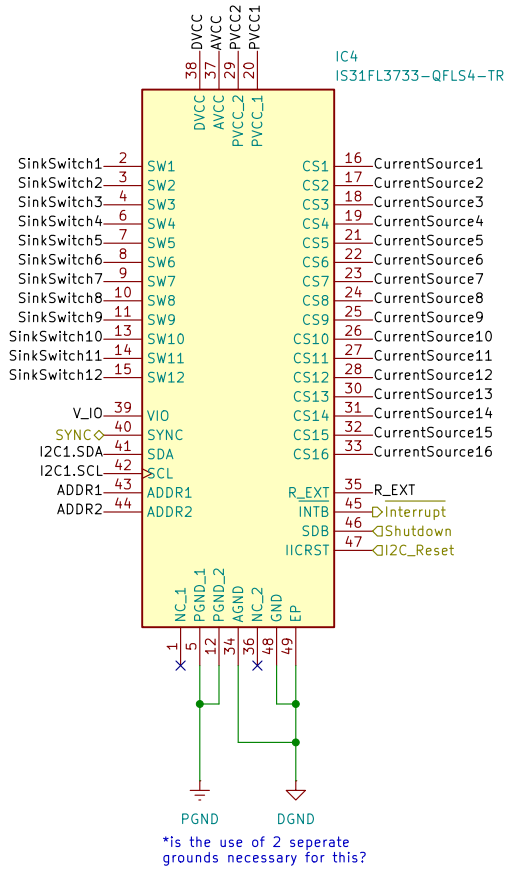
CurrentSource[1..16]
SinkSwitch[1..12]
I2C1{SDA, SCL}
SYNC
I2C_Reset
Shutdown
Interrupt

Blue
- SinkSwitch1
- SinkSwitch5
- SinkSwitch9

Green
- SinkSwitch2
- SinkSwitch6
- SinkSwitch10

Red
- SinkSwitch3
- SinkSwitch7
- SinkSwitch11

White
- SinkSwitch4
- SinkSwitch8
- SinkSwitch12



*is the use of 2 seperate grounds necessary for this?

Sheet: /LED Matricies/LED Matrix Controller 2/
File: matrix_controller_2.kicad_sch

Title: Slack

Size: A	Date: 2026-07-25	**Rev: A.0.5**
KiCad E.D.A. 10.0.5		Id: 7/7